Problem 1

Figure 1: Quanser's Aero 2; Quanser Inc.

The equations for this problem are taken from Quanser's Aero 2 two degree of freedom helicopter model. Quanser's Aero 2 is shown in Figure 1. The pitch is denoted by θ and the yaw by ψ.

The state for Quanser's two degree of freedom helicopter model is given by

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \theta \\ \psi \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}.$$

The linearized equations of motion for the helicopter about level flight admit a state space representation of the form

$$\dot{x} = Ax + Bu.$$

In this setting,

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -0.3208 & 0 & 0.0858 & 0 \\ 0 & 0 & 0 & -0.0806 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0.0232 & 0.0099 \\ -0.0224 & 0.0429 \end{bmatrix}$$

and

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}.$$

In this problem we are going to use an integral controller with two additional states

$$\dot{x}_5 = \theta(t) - \theta_d$$

$$\dot{x}_6 = \psi(t) - \psi_d$$

where θ_d and ψ_d are constants. Once we design our state feedback controller, we will replace θ_d and ψ_d by the reference signals $r_1(t)$ and $r_2(t)$ respectively.

If these reference signals do not change too fast, we hope to have

$$\theta(t) \to r_1(t), \qquad \psi(t) \to r_2(t).$$

Using this we obtain the augmented system

$$\dot{x} = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} x + \begin{bmatrix} B \\ 0 \end{bmatrix} u - \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}.$$

Explicitly,

$$\dot{x} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ -0.3208 & 0 & 0.0858 & 0 & 0 & 0 \\ 0 & 0 & 0 & -0.0806 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0.0232 & 0.0099 \\ -0.0224 & 0.0429 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} u - \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}.$$

We are not adding noise in this problem. However throughout the problem the initial conditions on the state and observer should be

$$\theta(0) = \pi, \qquad \psi(0) = \pi$$

and all other initial conditions are zero.

We assume that your observer can read the states x_1, x_2, x_5, and x_6. Therefore your observer gain L will map $\mathbb{C}^4$ into $\mathbb{C}^6$.

1. Let $r_1(t)$ be a square wave with amplitude 2 radians and period 12π. Let $r_2(t)$ be a sine wave with amplitude 2 radians and period 12π, that is

$$r_2(t) = 2\sin(t/6).$$

Use an integral observer based controller to design a state feedback controller

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = -K\hat{x}$$

such that $\theta(t)$ tracks the square wave $r_1(t)$ and $\psi(t)$ tracks the sine wave $r_2(t)$.

Note that your tracking of the sine wave can be off by a phase. However, in steady state $\psi(t)$ and $r_2(t)$ should have the same amplitude.

Problem 1

The equations for this problem are taken from Quanser's Aero 2 two degree of freedom helicopter model. The pitch is denoted by θ and the yaw by ψ.

Figure 2: Quanser Aero 2

The state vector is

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} \theta \\ \psi \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}$$

The linearized dynamics are

$$\dot{x} = Ax + Bu$$

where

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -0.3208 & 0 & 0.0858 & 0 \\ 0 & 0 & 0 & -0.0806 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0.0232 & 0.0099 \\ -0.0224 & 0.0429 \end{bmatrix}$$

and

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

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Integral Augmentation
To achieve tracking we introduce two integral states

$$\dot{x}_5 = \theta(t) - \theta_d$$

$$\dot{x}_6 = \psi(t) - \psi_d$$

The augmented state vector becomes

$$x_a = \begin{bmatrix} x \\ x_5 \\ x_6 \end{bmatrix}$$

The augmented system is

$$\dot{x}_a = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} x_a + \begin{bmatrix} B \\ 0 \end{bmatrix} u - \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$A_i = \begin{bmatrix} A & 0 \\ C & 0 \end{bmatrix} \qquad B_i = \begin{bmatrix} B \\ 0 \end{bmatrix}$$

$$C_i = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$W = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} \dot{x} \\ \dot{\hat{x}} \end{bmatrix} = \begin{bmatrix} A_i & -B_i K \\ LC_i & A_i - LC_i - B_i K \end{bmatrix} \begin{bmatrix} x \\ \hat{x} \end{bmatrix} + \begin{bmatrix} W \\ W \end{bmatrix} r$$

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Initial Conditions

The initial conditions used in simulation are

$$\theta(0) = \pi, \qquad \psi(0) = \pi$$

All other states are initialized to zero.

The observer is assumed to measure the states

$$x_1,\ x_2,\ x_5,\ x_6$$

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Controller Design
We use an integral observer based state feedback controller

$$u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = -K\hat{x}$$

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Part 1: Tracking Design
The reference signals are defined as

$$r_1(t) = \text{square wave with amplitude 2 and period } 12\pi$$

$$r_2(t) = 2\sin(t/6)$$

The objective is to design the controller such that

$$\theta(t) \to r_1(t)$$

$$\psi(t) \to r_2(t)$$

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Your Design

1. Choice of Weighting Matrices

Insert the weighting matrices used for the controller design.

$$Q = \begin{bmatrix} 300 & 0 & 0 & 0 & 0 & 0 \\ 0 & 300 & 0 & 0 & 0 & 0 \\ 0 & 0 & 30 & 0 & 0 & 0 \\ 0 & 0 & 0 & 30 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2000 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2000 \end{bmatrix}$$

$$R = \begin{bmatrix} 0.0100 & 0 \\ 0 & 0.0100 \end{bmatrix}$$

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2. State Feedback Gain

Insert the computed gain matrix

$$K = \begin{bmatrix} 393.52 & -168.87 & 176.65 & -57.22 & 402.00 & -195.94 \\ 195.28 & 354.37 & 89.88 & 124.60 & 195.94 & 402.00 \end{bmatrix}$$

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3. Observer Gain

Insert the observer gain

$$L = \begin{bmatrix} 6.4126 & 0 & 0.5708 & 0 \\ 0 & 6.2947 & 0 & 0.5639 \\ 5.7239 & 0 & 0.4160 & 0 \\ 0 & 4.9708 & 0 & 0.3849 \\ 0.5708 & 0 & 5.5512 & 0 \\ 0 & 0.5639 & 0 & 5.5507 \end{bmatrix}$$

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4. Closed Loop System

Write the closed loop system matrices used in MATLAB.

$$A_{cl} = \begin{bmatrix} 0 & 0 & 1.0000 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1.0000 & 0 & 0 \\ -11.3838 & 0.4094 & -4.9023 & 0.0939 & -11.2663 & 0.5660 \\ 0.4376 & -18.9853 & 0.1009 & -6.7078 & 0.5989 & -21.6350 \\ 1.0000 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1.0000 & 0 & 0 & 0 & 0 \end{bmatrix}$$

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5. Simulation Results

Insert the tracking plots for $\theta(t)$ and $\psi(t)$.

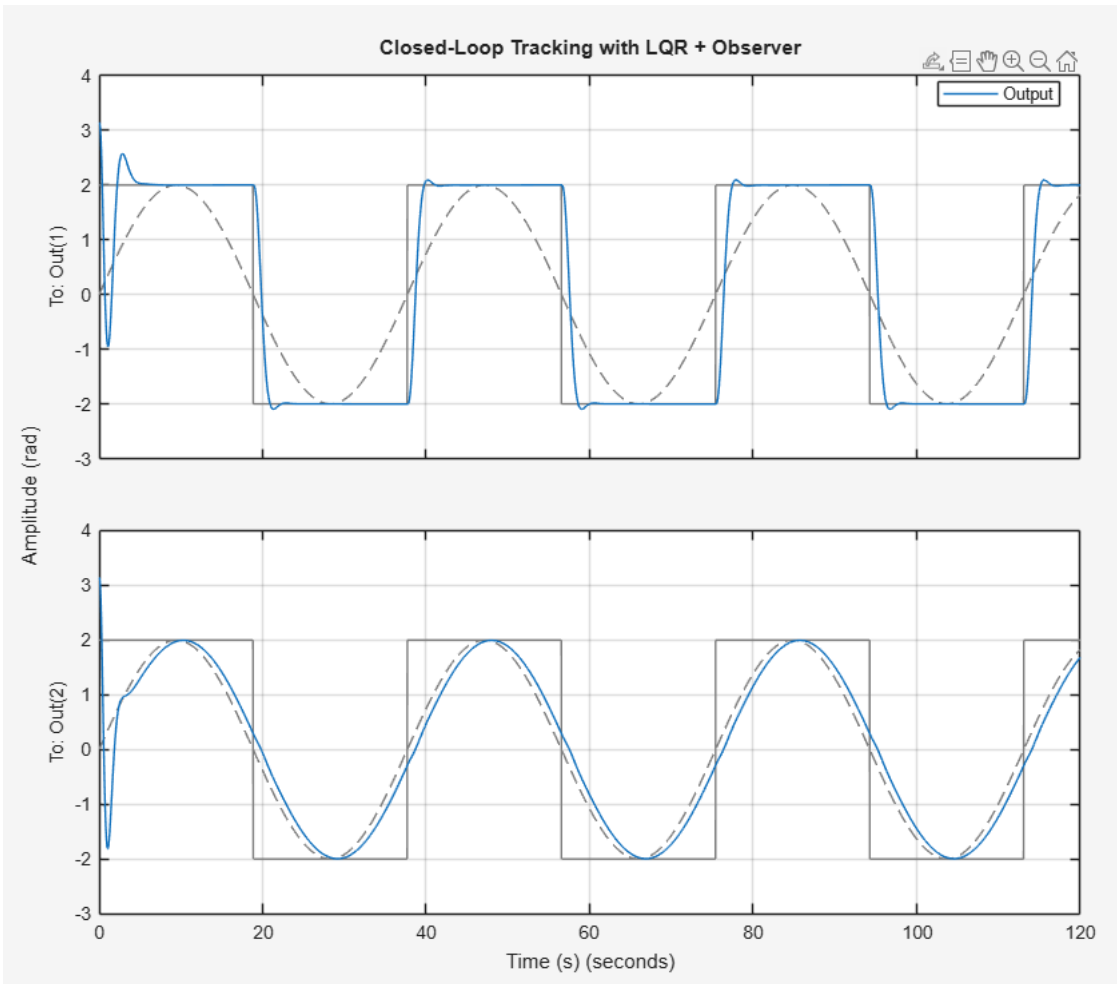

Figure 3: Tracking performance for square and sinusoidal reference signals

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Question 2

Plot $\theta(t)$ and $r_1(t)$ on the same graph for 120 seconds.

Solution

The reference signal is chosen as

$$r_1(t) = 2\,\mathrm{square}\left(\frac{t}{6}\right)$$

The closed-loop system is simulated using MATLAB with the LQR controller and observer designed in Question 1. The output $\theta(t)$ is compared with the reference signal $r_1(t)$.

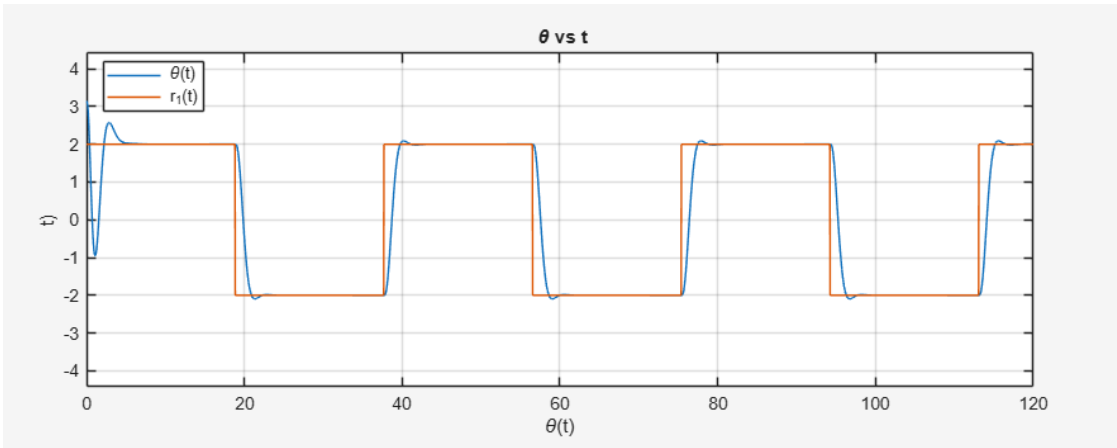

Figure 4: Tracking performance of $\theta(t)$ with respect to $r_1(t)$ over 120 seconds.

From the figure we observe that $\theta(t)$ closely tracks the square-wave reference signal with small transient error.

Question 3

Plot $\psi(t)$ and $r_2(t)$ on the same graph for 120 seconds.

Solution

The reference signal is defined as

$$r_2(t) = 2\sin\left(\frac{t}{6}\right)$$

The response $\psi(t)$ is obtained from the closed-loop simulation.

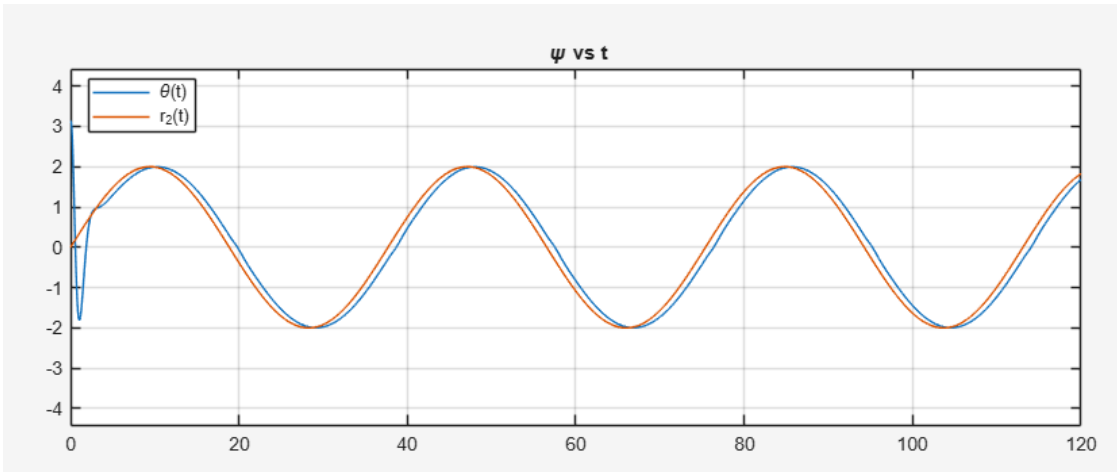

Figure 5: Tracking performance of $\psi(t)$ with respect to $r_2(t)$.

The output $\psi(t)$ follows the sinusoidal reference input with minimal steady-state error.

Question 4

Suppose that in steady state we want

$$\begin{bmatrix} \theta(t) \\ \psi(t) \end{bmatrix}$$

to trace a circle of radius 2 around the origin.

Solution

To generate a circular trajectory of radius 2, the reference signals are parameterized as

$$r_1(t) = 2\cos(\omega t)$$

$$r_2(t) = 2\sin(\omega t)$$

These references satisfy

$$\theta^2(t) + \psi^2(t) = 4$$

which represents a circle of radius 2 centered at the origin.

The closed-loop simulation is performed in MATLAB and the phase trajectory $[\theta(t), \psi(t)]^T$ is plotted.

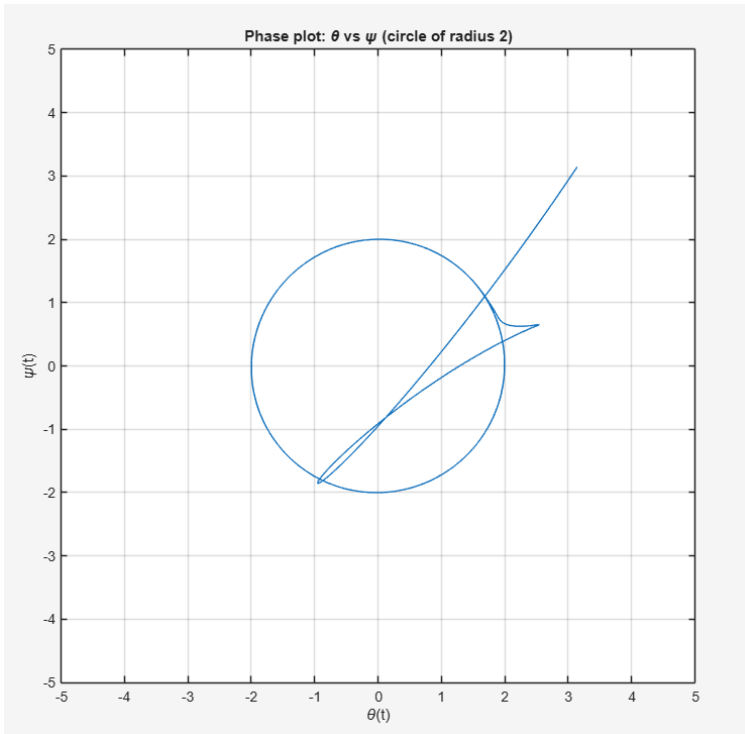

Figure 6: Phase trajectory showing circular motion of radius 2.

The trajectory approaches the desired circular path in steady state.

Question 5

Suppose that in steady state we want

$$\begin{bmatrix} \theta(t) \\ \psi(t) \end{bmatrix}$$

to trace a rotated ellipse around the origin.

Solution

An ellipse with semi-axes

$$a = 2, \quad b = \frac{1}{3}$$

can be parameterized as

$$x(t) = 2\cos(\omega t)$$

$$y(t) = \frac{1}{3}\sin(\omega t)$$

To rotate the ellipse by an angle $\phi = \frac{\pi}{4}$, the rotation matrix

$$R = \begin{bmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{bmatrix}$$

is applied:

$$\begin{bmatrix} r_1(t) \\ r_2(t) \end{bmatrix} = R \begin{bmatrix} 2\cos(\omega t) \\ \frac{1}{3}\sin(\omega t) \end{bmatrix}$$

Thus,

$$r_1(t) = \cos\phi \cdot x(t) - \sin\phi \cdot y(t)$$

$$r_2(t) = \sin\phi \cdot x(t) + \cos\phi \cdot y(t)$$

These reference signals generate a rotated elliptical trajectory.

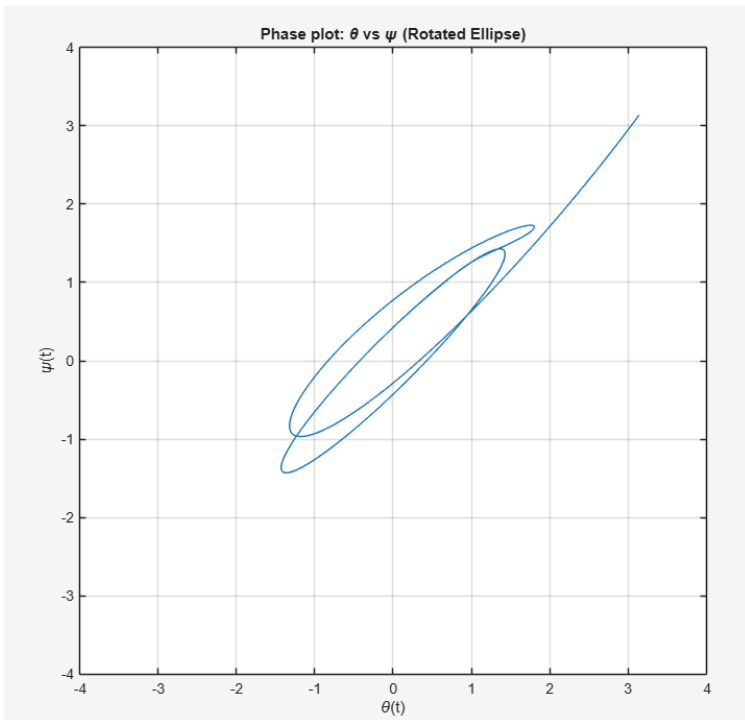

Figure 7: Rotated ellipse trajectory of $[\theta(t), \psi(t)]^T$.

The phase plot confirms that the system output converges to the desired rotated ellipse in steady state.